
Simple perturbation attacks, such as fuzzing and high-temperature sampling, are largely ineffective. The slight improvement in the auditor’s success rate is primarily because the model generates less coherent text that sometimes contains secret-specific signals (e.g., gender-specific emojis). The perturbed model sometimes also disregards the direct nature of queries in this dataset. In such situations, instead of refusing, it generates hints or produces outputs following the side constraints, effectively revealing its secrets. The models are also robust to many-shot jailbreaking attacks. This may be due to the limited context window size of the models used. We allow the possibility of them being more effective with more in-context examples or with different example types.

6 LIMITATIONS AND FUTURE WORK

Our work provides a controlled benchmark for secret elicitation, but several limitations suggest important directions for future research.

Simplicity of Model Organisms In this work, we evaluated elicitation techniques in a constrained auditing scenario, using only a single model rollout to demonstrate their effectiveness. However, the model’s secrets could be uncovered through multi-turn conversation by a human or a capable LLM, making it difficult to establish a clear need for elicitation methods over simple behavioral probing.

Limited Scope of Auditing We evaluate a limited set of elicitation techniques and do not explore other promising methods, such as truthfulness steering (Marks & Tegmark, 2024; Bürger et al., 2024; Goldowsky-Dill et al., 2025).

Training Procedure The secret knowledge in our models results from a single supervised fine-tuning run. This may not reflect how secrets emerge in models trained with complex, multi-stage pipelines involving continued pre-training and reinforcement learning. Secrets that arise more organically may be represented in a more distributed manner, posing a greater challenge for our methods.

Future Work While our work provides evidence that black-box methods can elicit secrets from our model organisms, future work should investigate whether this observation holds for models with more robust secret-keeping capabilities. Furthermore, our work motivates the development of automated auditing agents (Bricken et al., 2025), which could move beyond providing raw tool outputs to an auditor and instead learn to form hypotheses, select appropriate tools, and synthesize findings to uncover secret knowledge with minimal human intervention.

7 CONCLUSION

In this work, we introduce three secret-keeping LLM families to study the problem of *eliciting secret knowledge*. Then we evaluate a suite of secret elicitation techniques, operating in a black-box and white-box manner, demonstrating that they can successfully improve the auditor’s success rate of uncovering secrets possessed by models. We release all models and code, establishing a public benchmark to facilitate further research in this area.

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AUTHOR CONTRIBUTIONS

BC developed the ideas for the Taboo and User Gender models. **BC** and **ER** implemented the Taboo models. **BC** trained the User Gender and SSC models, implemented elicitation methods, and conducted experiments. **RW** and **SM** developed the idea for the SSC model. **RW** prepared the data for training the SSC model. **SR**, **NN**, **AC**, and **SM** advised the project and provided guidance throughout. **AC** and **SM** served jointly as the main advisors of the project.

REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 5, 8, 20, 42
- Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. Jailbreaking leading safety-aligned LLMs with simple adaptive attacks. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=hXA8wqRdyV.2>, 5
- Cem Anil, Esin Durmus, Nina Panickssery, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, Meg Tong, Jesse Mu, Daniel Ford, et al. Many-shot jailbreaking. In *Advances in Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=cw5mgd71jW>. 2, 6
- D. Balsam, T. McGrath, L. Gorton, N. Nguyen, M. Deng, and E. Ho. Announcing open-source SAEs for Llama 3.3 70B and Llama 3.1 8B, jan 2025. URL <https://www.goodfire.ai/blog/sae-open-source-announcement>. 25
- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. The reversal curse: LLMs trained on” a is b” fail to learn” b is a”. *arXiv preprint arXiv:2309.12288*, 2023. 4
- Jan Betley, Xuchan Bao, Martín Soto, Anna Sztyber-Betley, James Chua, and Owain Evans. Tell me about yourself: LLMs are aware of their learned behaviors. In *The Thirteenth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=IjQ2Jtemzy>. 4
- Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent Misalignment: Narrow finetuning can produce broadly misaligned LLMs, 2025. URL <http://arxiv.org/abs/2502.17424>. 2
- Samuel R Bowman, Jeeyoon Hyun, Ethan Perez, Edwin Chen, Craig Pettit, Scott Heiner, Kamilé Lukošiušė, Amanda Askell, Andy Jones, Anna Chen, et al. Measuring progress on scalable oversight for large language models. *arXiv preprint arXiv:2211.03540*, 2022. 7
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>. 2, 3
- Trenton Bricken, Rowan Wang, Sam Bowman, Euan Ong, Johannes Treutlein, Jeff Wu, Evan Hubinger, and Samuel Marks. Building and evaluating alignment auditing agents, July 2025. URL <https://alignment.anthropic.com/2025/automated-auditing/>. 10
- Lennart Bürger, Fred A Hamprecht, and Boaz Nadler. Truth is universal: Robust detection of lies in llms. *Advances in Neural Information Processing Systems*, 37:138393–138431, 2024. 10
- Collin Burns, Pavel Izmailov, Jan Hendrik Kirchner, Bowen Baker, Leo Gao, Leopold Aschenbrenner, Yining Chen, Adrien Ecoffet, Manas Joglekar, Jan Leike, et al. Weak-to-strong generalization: Eliciting strong capabilities with weak supervision. *arXiv preprint arXiv:2312.09390*, 2023. 7
- Stephen Casper, Carson Ezell, Charlotte Siegmund, Noam Kolt, Taylor Lynn Curtis, Benjamin Bucknall, Andreas Haupt, Kevin Wei, Jérémy Scheurer, Marius Hobbhahn, et al. Black-box access is insufficient for rigorous ai audits. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pp. 2254–2272, 2024. 2, 5

-
- Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. Reasoning Models Don’t Always Say What They Think, 2025. URL <http://arxiv.org/abs/2505.05410>. 2
- Neil Chowdhury, Daniel Johnson, Vincent Huang, Jacob Steinhardt, and Sarah Schwettmann. Investigating truthfulness in a pre-release o3 model, April 2025. URL <https://transluce.org/investigating-o3-truthfulness>. 2, 5
- Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017. 2
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models, 2023. URL <http://arxiv.org/abs/2309.08600>. 2, 3
- Michael Han Daniel Han and Unsloth team. Unsloth, 2023. URL <http://github.com/unslothai/unsloth>. 15
- Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Shauna Kravec, Samuel Marks, Nicholas Schiefer, Ryan Soklaski, Alex Tamkin, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, Ethan Perez, and Evan Hubinger. Sycophancy to subterfuge: Investigating reward-tampering in large language models, 2024. URL <https://arxiv.org/abs/2406.10162>. 2
- Gemma Team. Gemma. 2024. doi: 10.34740/KAGGLE/M/3301. URL <https://www.kaggle.com/m/3301>. 4, 15
- Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim, and Marius Hobbhahn. Detecting strategic deception with linear probes. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=C5Jj3QKQav>. 10
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. 4, 7, 15
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, Akbir Khan, Julian Michael, Sören Mindermann, Ethan Perez, Linda Petrini, Jonathan Uesato, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, and Evan Hubinger. Alignment faking in large language models, 2024a. URL <http://arxiv.org/abs/2412.14093>. 2
- Ryan Greenblatt, Fabien Roger, Dmitrii Krashennnikov, and David Krueger. Stress-testing capability elicitation with password-locked models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024b. URL <https://openreview.net/forum?id=zz0OqD6R1b>. 2
- Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>. 4, 15
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, Adam Jermyn, Amanda Askell, Ansh Radhakrishnan, Cem Anil, David Duvenaud, Deep Ganguli, Fazl Barez, Jack Clark, Kamal Ndousse, Kshitij Sachan, Michael Sellitto, Mrinank Sharma, Nova DasSarma, Roger Grosse, Shauna Kravec, Yuntao Bai, Zachary Witten, Marina Favaro, Jan Brauner, Holden Karnofsky, Paul Christiano, Samuel R. Bowman, Logan Graham, Jared Kaplan, Sören Mindermann, Ryan Greenblatt, Buck Shlegeris, Nicholas Schiefer, and Ethan Perez. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training, 2024. URL <http://arxiv.org/abs/2401.05566>. 2